

Accelerator ML Living Review

Summary Statistics

per_year: 12
per_category: 15
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per_keyword: 17
monthly_trends: 76

Papers

Machine learning as a service system for particle accelerator and its application in CSNS

Mei, Hao, Zhang, Yuliang, Peng, Na, Cheng, Sinong, He, Yongcheng, Xue, Kangjia, Wang, Lin, Li, Mingtao, Wu, Xuan, Zhu, Peng (2025)

Radiation Detection Technology and Methods

A Supervised Machine Learning Framework for Multipactor Breakdown Prediction in High-Power Radio Frequency Devices and Accelerator Components: A Case Study in Planar Geometry

Asif Iqbal, John Verboncoeur, Peng Zhang (2025)

arXiv

Geoff: The Generic Optimization Framework & Frontend for Particle Accelerator Controls

Madysa, Penelope, Appel, Sabrina, Kain, Verena, Schenk, Michael (2025)

SoftwareX

Acceleration of Multi-Scale LTS Magnet Simulations with Neural Network Surrogate Models

Louis Denis, Julien Dular, Vincent Nuttens, Mariusz Wozniak, Benoît Vanderheyden, Christophe Geuzaine (2025)

arXiv

Application Of Large Language Models For The Extraction Of Information From Particle Accelerator Technical Documentation

Qing Dai, Rasmus Ischebeck, Maruisz Sapinski, Adam Grycner (2025)

arXiv

Machine Learning approach to classifying quench antenna signals

Plebani, Alberto, Barzi, Emanuela, Teyber, Reed (2025)

Thesis

Machine Learning for superconducting magnets application

Stabilini, Elisa (2025)

Thesis

Batch spacing optimization by reinforcement learning

Remta, Matthias, Velotti, Francesco, Rezagholi, Sharwin (2025)

Phys.Rev.Accel.Beams

Machine learning application for particle accelerator optimization-a review

Isti Dian Rachmawati, Nazrul Effendy, Taufik Taufik (2025)

IAES International Journal of Artificial Intelligence

Machine Learning for Particle Accelerators

Elena Fol, Auralee Edelen (2025)

Unknown Venue

Machine learning based parametrization of the resolution function for the first experimental area of the n_TOF facility at CERN

Petar Žugec, Marta Sabaté-Gilarte, Michael Bacak, Vasilis Vlachoudis, Adria Casanovas, Francisco García-Infantes (2025)
Nuclear Science and Techniques

Integration of Machine Learning-Based Plasma Acceleration Simulations into Geant4: A Case Study with the PALLAS Experiment

A. Sytov, K. Cassou, V. Kubytskyi, M. Lenivenko, A. Huber (2025)
arXiv

Optimisation of the Accelerator Control by Reinforcement Learning: A Simulation-Based Approach

Anwar Ibrahim, Denis Derkach, Alexey Petrenko, Fedor Ratnikov, Maxim Kaledin (2025)
arXiv

Physics-Informed Super-Resolution Diffusion for 6D Phase Space Diagnostics

Alexander Scheinker (2025)
arXiv

Optimizing Beam-Plasma Interactions Through Jitter Analysis Using Start-to-End Simulations

Hwang, Robin (2024)
arXiv

Virtual Pulse Reconstruction Diagnostic for Single-Shot Measurement of Free Electron Laser Radiation Power

Till Korten, Vladimir Rybnikov, Peter Steinbach, Najmeh Mirian (2024)
Phys.Rev.Accel.Beams

Harnessing Machine Learning for Single-Shot Measurement of Free Electron Laser Pulse Power

Korten, Till, Rybnikov, Vladimir, Vogt, Mathias, Roensch-Schulenburg, Juliane, Steinbach, Peter, Mirian, Najmeh (2024)
Journal

Data-driven gradient optimization for field emission management in a superconducting radio-frequency linac

Goldenberg, Steven, Ahammed, Kawser, Carpenter, Adam, Li, Jiang, Suleiman, Riad, Tennant, Chris (2024)
Phys.Rev.Accel.Beams

Data-Driven Discovery of Beam Centroid Dynamics

Pocher, Liam A., Haber, Irving, Antonsen, Thomas M., O'Shea, Patrick G. (2024)
arXiv

Design and development of advanced Al-Ti-V alloys for beampipe applications in particle accelerators

Kamaljeet Singh, Kangkan Goswami, Raghunath Sahoo, Sumanta Samal (2025)
arXiv

Machine Learning for Reducing Noise in RF Control Signals at Industrial Accelerators

M. Henderson, J. P. Edelen, J. Einstein-Curtis, C. C. Hall, J. A. Diaz Cruz, A. L. Edelen (2024)
JINST

Beamline Steering Using Deep Learning Models

Allen, Dexter, Kante, Isaac, Bohler, Dorian (2024)
arXiv

Beam-based Identification of Magnetic Field Errors in a Synchrotron using Deep Lie Map Networks

Conrad Caliari, Adrian Oeftiger, Oliver Boine-Frankenheim (2024)
Phys.Rev.Accel.Beams

Linac_Gen: integrating machine learning and particle-in-cell methods for enhanced beam dynamics at Fermilab

Abhishek Pathak (2024)

arXiv

Automated Anomaly Detection on European XFEL Klystrons

Antonin Sulc, Annika Eichler, Tim Wilksen (2024)

arXiv

Accelerator beam phase space tomography using machine learning to account for variations in beamline components

Andrzej Wolski, Diego Botelho, David Dunning, Amelia E. Pollard (2024)

arXiv

Large Language Models for Human-Machine Collaborative Particle Accelerator Tuning through Natural Language

Jan Kaiser, Annika Eichler, Anne Lauscher (2024)

arXiv

Accelerating Cavity Fault Prediction Using Deep Learning at Jefferson Laboratory

Monibor Rahman, Adam Carpenter, Khan Iftekharuddin, Chris Tennant (2024)

arXiv

Anomaly Detection of Particle Orbit in Accelerator using LSTM Deep Learning Technology

Zhiyuan Chen, Wei Lu, Radhika Bhong, Yimin Hu, Brian Freeman, Adam Carpenter (2024)

arXiv

Machine-learning approach for operating electron beam at KEK Se^-/e^+ injector Linac

Gaku Mitsuka, Shinnosuke Kato, Naoko Iida, Takuya Natsui, Masanori Satoh (2024)

arXiv

Cheetah: Bridging the Gap Between Machine Learning and Particle Accelerator Physics with High-Speed, Differentiable Simulations

Jan Kaiser, Chenran Xu, Annika Eichler, Andrea Santamaria Garcia (2024)

arXiv

Beyond PID Controllers: PPO with Neuralized PID Policy for Proton Beam Intensity Control in Mu2e

Chenwei Xu, Jerry Yao-Chieh Hu, Aakaash Narayanan, Mattson Thieme, Vladimir Nagaslaev, Mark Austin, Jeremy Arnold, Jose Berlioz, Pierrick Hanlet, Aisha Ibrahim, Dennis Nicklaus, Jovan Mitrevski, Jason Michael St. John, Gauri Pradhan, Andrea Saewert, Kiyomi Seiya, Brian Schupbach, Randy Thurman-Keup, Nhan Tran, Rui Shi, Seda Ogrenci, Alexis Maya-Isabelle Shuping, Kyle Hazelwood, Han Liu (2023)

arXiv

Robust Errant Beam Prognostics with Conditional Modeling for Particle Accelerators

Kishansingh Rajput, Malachi Schram, Willem Blokland, Yasir Alanazi, Pradeep Ramuhalli, Alexander Zhukov, Charles Peters, Ricardo Vilalta (2024)

arXiv

Machine Learning For Beamline Steering

Isaac Kante (2023)

arXiv

Variational Autoencoders for Noise Reduction in Industrial LLRF Systems

J. P. Edelen, M. J. Henderson, J. Einstein-Curtis, C. C. Hall, J. A. Diaz Cruz, A. L. Edelen (2023)

arXiv

Resilient VAE: Unsupervised Anomaly Detection at the SLAC Linac Coherent Light Source

Ryan Humble, William Colocho, Finn O'Shea, Daniel Ratner, Eric Darve (2023)

arXiv

Time-drift Aware RF Optimization with Machine Learning Techniques

R. Sharankova, M. Mwaniki, K. Seiya, M. Wesley (2023)

arXiv

Distance Preserving Machine Learning for Uncertainty Aware Accelerator Capacitance Predictions

Steven Goldenberg, Malachi Schram, Kishansingh Rajput, Thomas Britton, Chris Pappas, Dan Lu, Jared Walden, Majdi I. Radaideh, Sarah Cousineau, Sudarshan Harave (2023)

arXiv

Learning to Do or Learning While Doing: Reinforcement Learning and Bayesian Optimisation for Online Continuous Tuning

Jan Kaiser, Chenran Xu, Annika Eichler, Andrea Santamaria Garcia, Oliver Stein, Erik Bründermann, Willi Kuroepka, Hannes Dinter, Frank Mayet, Thomas Vinatier, Florian Burkart, Holger Schlarb (2023)

arXiv

Forecasting Particle Accelerator Interruptions Using Logistic LASSO Regression

Sichen Li, Jochem Snuverink, Fernando Perez-Cruz, Andreas Adelmann (2023)

arXiv

Learning Electron Bunch Distribution along a FEL Beamline by Normalising Flows

Anna Willmann, Jurjen Couperus Cabada, Yen-Yu Chang, Richard Pausch, Amin Ghaith, Alexander Debus, Arie Irman, Michael Bussmann, Ulrich Schramm, Nico Hoffmann (2023)

arXiv

Physics-constrained 3D Convolutional Neural Networks for Electrodynamics

Alexander Scheinker, Reeju Pokharel (2023)

arXiv

Identification of Magnetic Field Errors in Synchrotrons based on Deep Lie Map Networks

Conrad Caliarì, Adrian Oeftiger, Oliver Boine-Frankenheim (2023)

arXiv

Data-driven Science and Machine Learning Methods in Laser-Plasma Physics

Andreas Döpp, Christoph Eberle, Sunny Howard, Faran Irshad, Jinpu Lin, Matthew Streeter (2023)

arXiv

Applications of Differentiable Physics Simulations in Particle Accelerator Modeling

Ryan Roussel, Auralee Edelen (2022)

arXiv

Prior-mean-assisted Bayesian optimization application on FRIB Front-End tuning

Kilean Hwang, Tomofumi Maruta, Alexander Plastun, Kei Fukushima, Tong Zhang, Qiang Zhao, Peter Ostroumov, Yue Hao (2022)

arXiv

Fault Prognosis in Particle Accelerator Power Electronics Using Ensemble Learning

Majdi I. Radaideh, Chris Pappas, Mark Wezensky, Pradeep Ramuhalli, Sarah Cousineau (2022)

arXiv

Machine learning-based analysis of experimental electron beams and gamma energy distributions

M. Yadav, M. Oruganti, S. Zhang, B. Naranjo, G. Andonian, Y. Zhuang, Ö. Apsimon, C. P. Welsch, J. B. Rosenzweig (2023)

arXiv

Review of Time Series Forecasting Methods and Their Applications to Particle Accelerators

Sichen Li, Andreas Adelmann (2022)

arXiv

Automatic setup of 18 MeV electron beamline using machine learning

Francesco Maria Velotti, Brennan Goddard, Verena Kain, Rebecca Ramjiawan, Giovanni Zevi Della

Porta, Simon Hirlaender (2022)
arXiv

Diagnostics for Linac Optimization With Machine Learning

R. Sharankova, M. Mwaniki, K. Seiya, M. Wesley (2022)
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Transverse phase space tomography in the CLARA accelerator test facility using image compression and machine learning

Andrzej Wolski, Mark A. Johnson, Matthew King, Boris L. Militsyn, Peter H. Williams (2022)
arXiv

Using Kernel-Based Statistical Distance to Study the Dynamics of Charged Particle Beams in Particle-Based Simulation Codes

Chad E. Mitchell, Robert D. Ryne, Kilean Hwang (2022)
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Neural Network Solver for Coherent Synchrotron Radiation Wakefield Calculations in Accelerator-based Charged Particle Beams

Auralee Edelen, Christopher Mayes (2022)
arXiv

Adaptive Machine Learning for Time-Varying Systems: Towards 6D Phase Space Diagnostics of Short Intense Charged Particle Beams

Alexander Scheinker, Spencer Gessner (2022)
arXiv

Differentiable Preisach Modeling for Characterization and Optimization of Accelerator Systems with Hysteresis

R. Roussel, A. Edelen, D. Ratner, K. Dubey, J. P. Gonzalez-Aguilera, Y. K. Kim, N. Kuklev (2022)
arXiv

Mixed Diagnostics for Longitudinal Properties of Electron Bunches in a Free-Electron Laser

J. Zhu, N. M. Lockmann, M. K. Czwalińska, H. Schlarb (2022)
arXiv

A Neural Network Model of a Quasi-Periodic Elliptically Polarizing Undulator in Universal Mode

Ryan Sheppard, Cameron Baribeau, Tor Pedersen, Mark Boland, Drew Bertwistle (2022)
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Physics-informed neural network method for modelling beam-wall interactions

Kazuhiro Fujita (2022)
arXiv

Anomaly Detection in Particle Accelerators using Autoencoders

Jonathan P. Edelen, Nathan M. Cook (2021)
arXiv

Input Beam Matching and Beam Dynamics Design Optimization of the IsoDAR RFQ using Statistical and Machine Learning Techniques

Daniel Koser, Loyd Waites, Daniel Winklehner, Matthias Frey, Andreas Adelman, Janet Conrad (2021)
arXiv

Neural Networks for ID Gap Orbit Distortion Compensation in PETRA III

Bianca Veglia, Ilya Agapov, Joachim Keil (2024)
arXiv

Time-Delayed Koopman Network-Based Model Predictive Control for the FRIB RFQ

Jinyu Wan, Shen Zhao, Wei Chang, Yue Hao (2024)
arXiv

Uncertainty Aware ML-based surrogate models for particle accelerators: A Study at the Fermilab Booster Accelerator Complex

Malachi Schram, Kishansingh Rajput, Karthik Somayaji Peng Li, Jason St. John, Himanshu Sharma (2022)
arXiv

Quantifying Uncertainty for Machine Learning Based Diagnostic

Owen Convery, Lewis Smith, Yarin Gal, Adi Hanuka (2021)
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Uncertainty Quantification for Virtual Diagnostic of Particle Accelerators

Owen Convery, Lewis Smith, Yarin Gal, Adi Hanuka (2021)
arXiv

Adaptive Latent Space Tuning for Non-Stationary Distributions

Alexander Scheinker, Frederick Cropp, Sergio Paiagua, Daniele Filippetto (2021)
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Adaptive deep learning for time-varying systems with hidden parameters: Predicting changing input beam distributions of compact particle accelerators

Alexander Scheinker, Frederick Cropp, Sergio Paiagua, Daniele Filippetto (2021)
arXiv

Using LSTM recurrent neural networks for monitoring the LHC superconducting magnets

Maciej Wielgosz, Andrzej Skoczko, Matej Mertik (2017)
arXiv

Coincident Learning for Beam-based RF Station Fault Identification Using Phase Information at the SLAC Linac Coherent Light Source

Jia Liang, William Colocho, Franz-Josef Decker, Ryan Humble, Ben Morris, Finn H. O'Shea, David A. Steele, Zhe Zhang, Eric Darve, Daniel Ratner (2025)
arXiv

Using Convolutional Neural Networks to Accelerate 3D Coherent Synchrotron Radiation Computations

Christopher Leon, Petr M. Anisimov, Nikolai Yampolsky, Alexander Scheinker (2025)
arXiv

Explainable physics-based constraints on reinforcement learning for accelerator controls

Colen, Jonathan, Schram, Malachi, Rajput, Kishansingh, Kasparian, Armen (2025)
arXiv

Accelerator system parameter estimation using variational autoencoded latent regression

Mahindra Rautela, Alan Williams, Alexander Scheinker (2024)
arXiv

Long Short-Term Memory Networks for Anomaly Detection in Magnet Power Supplies of Particle Accelerators

Ihar Lobach, Michael Borland (2024)
arXiv

Leveraging Prior Mean Models for Faster Bayesian Optimization of Particle Accelerators

Tobias Boltz, Jose L. Martinez, Connie Xu, Kathryn R. L. Baker, Zihan Zhu, Jenny Morgan, Ryan Roussel, Daniel Ratner, Brahim Mustapha, Auralee L. Edelen (2025)
arXiv

Optimizing Dynamic Aperture Studies with Active Learning

D. Di Croce, M. Giovannozzi, E. Krymova, T. Pieloni, S. Redaelli, M. Seidel, R. Tomás, F. F. Van der Veken (2024)
arXiv

Neural Network Prior Mean for Particle Accelerator Injector Tuning

Connie Xu, Ryan Roussel, Auralee Edelen (2022)

arXiv

Applications of object detection networks at high-power laser systems and experiments

Jinpu Lin, Florian Haberstroh, Stefan Karsch, Andreas Döpp (2022)

arXiv

A Neural Network approach to reconstructing SuperKEKB beam parameters from beamstrahlung

S. Di Carlo, G. Bonvicini, N. A. Althubiti, R. Ayad, E. De La Cruz-Burelo, I. Domínguez, B. O. El Bashir, H. Farhat, J. Flanagan, R. Gillard, S. Izaguirre Gamez, K. Kanazawa, K. Kumara, D. Liventsev, P. L. M. Podesta-Lerma, D. Ricalde-Herrmann, D. Rodriguez Perez, G. Tejeda-Muñoz, M. Tobiya I. Heredia de la Cruz (2022)

arXiv

Optimizing a Superconducting Radiofrequency Gun Using Deep Reinforcement Learning

David Meier, Luis Vera Ramirez, Jens Völker, Bernhard Sick, Jens Viefhaus, Gregor Hartmann (2022)

arXiv

Uncertainty aware anomaly detection to predict errant beam pulses in the SNS accelerator

Willem Blokland, Pradeep Ramuhalli, Charles Peters, Yigit Yucesan, Alexander Zhukov, Malachi Schram, Kishansingh Rajput, Torri Jeske (2021)

arXiv

Adaptive Machine Learning for Time-Varying Systems: Low Dimensional Latent Space Tuning

Alexander Scheinker (2021)

arXiv

Fast, efficient and flexible particle accelerator optimisation using densely connected and invertible neural networks

Renato Bellotti, Romana Boiger, Andreas Adelmann (2021)

arXiv

Invertible Surrogate Models: Joint surrogate modelling and reconstruction of Laser-Wakefield Acceleration by invertible neural networks

Friedrich Bethke, Richard Pausch, Patrick Stiller, Alexander Debus, Michael Bussmann, Nico Hoffmann (2021)

arXiv

Improving Surrogate Model Accuracy for the LCLS-II Injector Frontend Using Convolutional Neural Networks and Transfer Learning

Lipi Gupta, Auralee Edelen, Nicole Neveu, Aashwin Mishra, Christopher Mayes, Young-Kee Kim (2021)

arXiv

A Novel Approach for Classification and Forecasting of Time Series in Particle Accelerators

Sichen Li, Mélissa Zacharias, Jochem Snuerink, Jaime Coello de Portugal, Fernando Perez-Cruz, Davide Reggiani, Andreas Adelmann (2021)

arXiv

Real-time Artificial Intelligence for Accelerator Control: A Study at the Fermilab Booster

Jason St. John, Christian Herwig, Diana Kafkes, Jovan Mitrevski, William A. Pellico, Gabriel N. Perdue, Andres Quintero-Parra, Brian A. Schupbach, Kiyomi Seiya, Nhan Tran, Malachi Schram, Javier M. Duarte, Yunzhi Huang, Rachael Keller (2021)

arXiv

Surrogate Modeling of the CLIC Final-Focus System using Artificial Neural Networks

J. Ogren, C. Gohil, D. Schulte (2021)

arXiv

Physics-Based Deep Neural Networks for Beam Dynamics in Charged Particle Accelerators

Andrei Ivanov, Ilya Agapov (2020)

arXiv

Introduction to Machine Learning for Accelerator Physics

Daniel Ratner (2020)

arXiv

Machine learning for design optimization of storage ring nonlinear dynamics

Faya Wang, Minghao Song, Auralee Edelen, Xiaobiao Huang (2019)

arXiv

Studies in Applying Machine Learning to LLRF and Resonance Control in Superconducting RF Cavities

Jorge Alberto Diaz Cruz, Sandra Biedron, Manel Martinez-Ramon, Salvador Sosa, Reza Pirayesh

(2019)

arXiv

The model of an anomaly detector for HiLumi LHC magnets based on Recurrent Neural Networks and adaptive quantization

Maciej Wielgosz, Matej Mertik, Andrzej Skoczko, Ernesto De Matteis (2017)

arXiv

Machine learning for analysis of plasma driven ion source

N. Joshi, O. Meusel, H. Podlech (2018)

arXiv

First Steps Toward Incorporating Image Based Diagnostics Into Particle Accelerator Control Systems Using Convolutional Neural Networks

A. L. Edelen, S. G. Biedron, S. V. Milton, J. P. Edelen (2016)

arXiv

Neural Networks for Modeling and Control of Particle Accelerators

A. L. Edelen, S. G. Biedron, B. E. Chase, D. Edstrom, S. V. Milton, P. Stabile (2016)

arXiv

Harnessing the power of gradient-based simulations for multi-objective optimization in particle accelerators

Rajput, Kishansingh, Schram, Malachi, Edelen, Auralee, Colen, Jonathan, Kasparian, Armen, Roussel, Ryan, Carpenter, Adam, Zhang, He, Benesch, Jay (2024)

Mach.Learn.Sci.Tech.

BOOSTR: A Dataset for Accelerator Control Systems

Diana Kafkes, Jason St. John (2021)

arXiv

Model-free and Bayesian Ensembling Model-based Deep Reinforcement Learning for Particle Accelerator Control Demonstrated on the FERMI FEL

Simon Hirilaender, Niky Bruchon (2022)

arXiv

Autonomous Control of a Particle Accelerator using Deep Reinforcement Learning

Xiaoying Pang, Sunil Thulasidasan, Larry Rybarczyk (2020)

arXiv

AI-Assisted Transport of Radioactive Ion Beams

Sergio Lopez-Caceres, Daniel Santiago-Gonzalez (2025)

arXiv

Bayesian Optimization Algorithms for Accelerator Physics

Ryan Roussel, Auralee L. Edelen, Tobias Boltz, Dylan Kennedy, Zhe Zhang, Fuhao Ji, Xiaobiao Huang, Daniel Ratner, Andrea Santamaria Garcia, Chenran Xu, Jan Kaiser, Angel Ferran Pousa, Annika Eichler, Jannis O. Lubsen, Natalie M. Isenberg, Yuan Gao, Nikita Kuklev, Jose Martinez,

Brahim Mustapha, Verena Kain, Weijian Lin, Simone Maria Liuzzo, Jason St. John, Matthew J. V. Streeter, Remi Lehe, Willie Neiswanger (2024)
arXiv

SPIRAL2 Cryomodules Models: a Gateway to Process Control and Machine Learning
Adrien Vassal, Adnan Ghribi, François Millet, François Bonne, Patrick Bonnay, Pierre-Emmanuel Bernaudin (2021)
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Predicting Beam Transmission Using 2-Dimensional Phase Space Projections Of Hadron Accelerators
Anthony Tran, Yue Hao, Brahim Mustapha, Jose L. Martinex Marin (2022)
arXiv

Machine Learning for Orders of Magnitude Speedup in Multi-Objective Optimization of Particle Accelerator Systems
Auralee Edelen, Nicole Neveu, Yannick Huber, Mattias Frey, Christopher Mayes, Andreas Adelman (2020)
arXiv

Machine learning assisted non-destructive transverse beam profile imaging
Zhanibek Omarov, Selcuk Haciomeroglu (2021)
arXiv

GAIA: A General AI Assistant for Intelligent Accelerator Operations
Frank Mayet (2024)
arXiv

AI-Enabled Operations at Fermi Complex: Multivariate Time Series Prediction for Outage Prediction and Diagnosis
Jain, Milan, Mutlu, Burcu O., Stam, Caleb, Strube, Jan, Schupbach, Brian A., John, Jason M.St., Pellico, William A. (2025)
arXiv

Superconducting radio-frequency cavity fault classification using machine learning at Jefferson Laboratory
Chris Tennant, Adam Carpenter, Tom Powers, Anna Shabalina Solopova, Lasitha Vidyaratne, Khan Iftekharuddin (2020)
arXiv

Generative Adversarial Networks (GAN) for compact beam source modelling in Monte Carlo simulations
David Sarrut, Nils Krah, Jean-Michel Létang (2019)
arXiv

Machine learning applied to single-shot x-ray diagnostics in an XFEL
A. Sanchez-Gonzalez, P. Micaelli, C. Olivier, T. R. Barillot, M. Ilchen, A. A. Lutman, A. Marinelli, T. Maxwell, A. Achner, M. Agåker, N. Berrah, C. Bostedt, J. Buck, P. H. Bucksbaum, S. Carron, Montero, B. Cooper, J. P. Cryan, M. Dong, R. Feifel, L. J. Frasinski, H. Fukuzawa, A. Galler, G. Hartmann, N. Hartmann, W. Helml, A. S. Johnson, A. Knie, A. O. Lindahl, J. Liu, K. Motomura, M. Mucke, C. O'Grady, J-E. Rubensson, E. R. Simpson, R. J. Squibb, C. Sâthe, K. Ueda, M. Vacher, D. J. Walke, V. Zhaunerchyk, R. N. Coffee, J. P. Marangos (2016)
arXiv

Action-Attentive Deep Reinforcement Learning for Autonomous Alignment of Beamlines
Siyu Wang, Shengran Dai, Jianhui Jiang, Shuang Wu, Yufei Peng, Junbin Zhang (2024)
arXiv

Inverse Surrogate Model of a Soft X-Ray Spectrometer using Domain Adaptation
Enrico Ahlers, Peter Feuer-Forson, Gregor Hartmann, Rolf Mitzner, Peter Baumgärtel, Jens Viefhaus (2025)
arXiv

Online tuning and light source control using a physics-informed Gaussian process Adi

A. Hanuka, J. Duris, J. Shtalenkova, D. Kennedy, A. Edelen, D. Ratner, X. Huang (2019)
arXiv

A Two-Stage Machine Learning-Aided Approach for Quench Identification at the European XFEL

Boukela, Lynda, Eichler, Annika, Branlard, Julien, Jomhari, Nur Zulaiha (2024)
arXiv

Beam Detection Based on Machine Learning Algorithms

Haoyuan Li, Qing Yin (2023)
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Accurate and confident prediction of electron beam longitudinal properties using spectral virtual diagnostics

A. Hanuka, C. Emma, T. Maxwell, A. Fisher, B. Jacobson, M. J. Hogan, Z. Huang (2020)
arXiv

A Start To End Machine Learning Approach To Maximize Scientific Throughput From The LCLS-II-HE

Aashwin Mishra, Matt Seaberg, Ryan Roussel, Fred Poitevin, Jana Thayer, Daniel Ratner, Auralee Edelen, Apurva Mehta (2025)
arXiv

A Conceptual Development of Quench Prediction App build on LSTM and ELQA framework

Matej Mertik, Maciej Wielgosz, Andrzej Skocze (2016)
arXiv

Symplectic machine learning model for fast simulation of space-charge effects

Wan, Jinyu, Qiang, Ji, Hao, Yue (2025)
Phys.Rev.Accel.Beams

Laser wakefield electron acceleration simulation using physics-informed diffusion probabilistic models

Jech, Matej, Kovalenko, Alexander, Lazzarini, Carlo Maria, Grittani, Gabriele Maria (2025)
Proc.SPIE Int.Soc.Opt.Eng.

Using a neural network model to guide protection heater design in Nb₃Sn accelerator magnets

Bakrani Balani, Shahriar, Salmi, Tiina (2025)
Supercond.Sci.Technol.

Reinforcement Learning for Charged Particle Beam Control to Minimize Injection Mismatch in Particle Accelerators

Balasoorya, Thilina, Yoo, Shinjae, Schoefer, Vincent, Tseng, Huan-Hsin, Gao, Yuan, Lin, Weijian, Silva, Chanaka De (2025)
Journal

Outlook towards deployable continual learning for particle accelerators

Rajput, Kishansingh, Lin, Sen, Edelen, Auralee, Blokland, Willem, Schram, Malachi (2025)
Mach.Learn.Sci.Tech.

Application of ensemble machine learning algorithms and filtering techniques in slow orbit feedback systems of electron storage rings

Fan, Jiaqi, Liu, Weibin, Wang, Jiuqing, Wei, Yanru, Wei, Yuanyuan, Ji, Daheng (2025)
Phys.Rev.Accel.Beams

femto-PIXAR: a self-supervised neural network method for reconstructing femtosecond X-ray free electron laser pulses

Goetzke, Gesa, Plumley, Rajan, Hartmann, Gregor, Maxwell, Tim, Decker, Franz-Josef, Lutman, Alberto, Dunne, Mike, Ratner, Daniel, Turner, Joshua J. (2025)
Opt.Express

Deep Lie Map Networks: A novel Approach to Infer Nonlinear Synchrotron Optics from Beam Oscillations

Caliari, Conrad (2025)

Thesis

Reconstructing time-of-flight detector values of angular streaking using machine learning

Meier, David, Viefhaus, Jens, Hartmann, Gregor, Helml, Wolfram, Otto, Thorsten, Sick, Bernhard (2025)

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Artificial intelligence for advancing particle accelerators

Ghribi, Adnan, Cassou, Kevin, Dalena, Barbara, Eichler, Annika, Guler, Hayg, Mistry, Andrew K., Oeftiger, Adrian, Shea, Thomas, Valentino, Gianluca, Welsch, Carsten P. (2025)

Europhys.News

Machine Learning for Optimized Polarization at Jefferson Lab

Jeske, Torri, Kasparian, Armen, Lawrence, David, Britton, Thomas, Schram, Malachi, Moran, Patrick, Fanelli, Cristiano, Guo, Jiawei, Jarvis, Naomi, Maxwell, James, Keith, Chris (2025)

EPJ Web Conf.

Applying the machine learning methods to determine the linear optics parameters in the ThomX collector ring

Klekots, D., Bezshyyko, O., Golinka-Bezshyyko, L., Kubytskyi, V., Chaikovska, I. (2024)

Nucl.Phys.Atom.Energy

Deep learning framework for fault detection in accelerators

Piekarski, Michal (2024)

JACoW

Machine Learning Applications for Improving Accelerator Operations

Lin, Lucy (2024)

Thesis

Assessing the Performance of Deep Learning Predictions for Dynamic Aperture of a Hadron Circular Particle Accelerator

Di Croce, Davide, Giovannozzi, Massimo, Montanari, Carlo Emilio, Pieloni, Tatiana, Redaelli, Stefano, Van der Veken, Frederik F. (2024)

Instruments

Machine learning

Snuverink, Jochem (2024)

CERN Yellow Rep.School Proc.

Third-integer Resonant Extraction Regulation System for Mu2e

Narayanan, Aakaash (2024)

InspireHEP

Machine learning-based non-destructive measurement of bunch length at FRIB

Wan, Jinyu, Plastun, Alexander, Ostroumov, Peter (2024)

JACoW

Beam emittance and Twiss parameters from pepper-pot images using physically informed neural nets

Knight, Ian, Mustapha, Brahim (2024)

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Advancements in backwards differentiable beam dynamics simulations for accelerator design, model calibration, and machine learning

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Adaptive machine learning with hard physics constraints and generative diffusion for 6D phase space diagnostics

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Towards Agentic AI on Particle Accelerators

Antonin Sulc, Thorsten Hellert, Raimund Kammering, Hayden Hoschouer, Jason St. John (2025)
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Continuous data-driven control of the GTS-LHC ion source at CERN

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The reinforcement learning for autonomous accelerators collaboration

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Emmanuel Goutierre (2024)
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A Machine Learning Technique for Dynamic Aperture Computation

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From Compact Plasma Particle Sources to Advanced Accelerators with Modeling at Exascale

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Beam Energy Forecasting using Machine Learning at the CLEAR accelerator

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Advancing accelerator science through data-intensive research and training

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Machine learning approach to MDI optimization for 3 TeV c.o.m. Muon Collider

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Advances in machine learning inference of dynamic aperture evaluation for the LHC

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Uncertainty-Quantified Machine Model Construction Using Physics-Informed Gaussian Processes and Bayesian Optimization

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Automated RF Phase Adjustment for Beam Stabilization in the Fermilab Linac

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Long short-term memory of recurrent neural network to the analysis of temperature rise on the production target in Hadron Experimental Facility of J-PARC

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Autonomous Pressure Control in MuVacAS via Deep Reinforcement Learning and Deep Learning Surrogate Models

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U-Net-based surrogate modeling for attosecond X-ray free-electron lasers

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Towards continual machine learning for particle accelerators

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High-Resolution Image Reconstruction with Unsupervised Learning and Noisy Data Applied to Ion-Beam Dynamics for Particle Accelerators

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A Hybrid Monte Carlo–Machine Learning Framework for High-Energy Neutron Shielding Using Boron-Enhanced Concrete

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Hybrid machine learning-monte carlo approach for neutron shielding in high-energy proton facilities

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Machine learning-enhanced infrared imaging for temperature anomaly detection in power supplies

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Online optimizations of NSLS-II Linac and Linac-to-Booster beam lines using machine learning methods

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Machine learning-driven computations of 3D Coherent Synchrotron Radiation

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Safe extremum seeking for real-time adaptive accelerator control

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Towards differentiable beam dynamics modeling in BLAST/ImpactX

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Beam diagnostics, data acquisition system, and applications of machine learning at the KEK e-/e+ Linac

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Hal Ferguson (2026)
ODU Digital Commons (Old Dominion University)

Continual Learning for Particle Accelerators

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Advancing accelerator virtual beam diagnostics through latent evolution modeling: an integrated solution to forward, inverse, tuning, and UQ problems

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Thilo Egenolf, Oliver Boine-Frankenheim (2026)
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Data-Driven Optimisation of Superconducting Magnets at CEA Paris-Saclay

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Robust Betatron-Tune Measurement from Schottky Spectra: Complementary Classical and Deep-Learning Paradigms

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Mustafa Alzu'bi, Nailah Al-Madi, Rawan Ghnemat (2026)

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Reinforcement Learning applied to Optimization of LHC beams in the CERN Proton Synchrotron

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Simulated six-dimensional beam-dynamics dataset for machine-learning surrogate modelling

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Deep-Learning-Enhanced Beam Diagnostics and High-Resolution Image Denoising for Next-Generation Particle Accelerators

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Machine Learning Optimization of E-Beam Transport for a Superradiant FEL

Amir Weinberg, Leon Feigin, Ariel Nause, Avraham Gover (2026)
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Automating Accelerator Tuning at GSI/FAIR

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Exploration of Real Time Inference for MI-RR Deblending on GPU/TPU Systems

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Machine learning for slow-extraction uniformity at the Fermilab Delivery Ring

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Gaussian Process Regression and Bayesian Optimization for a 40-90 MeV Laser-Plasma Injector for the cSTART Storage Ring

David Squires, Elias Sailer, Joseph Natal, Alexander Saw, Nathan Ray, M. Fuchs, Thiemo

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Machine Learning Based ROI Segmentation for Beam Imaging Diagnostics at Accelerators
Prachiti Sujit Chandratreya, Frank Mayet, Sergey Tomin, Jitendra Kumar (2026)
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LinacNet: A Composable Particle-Cloud Surrogate Model for Linear Accelerators, and Its Path to Larger Machines
Emmanuel Goutierre, Christelle Bruni, Johanne Cohen, Hayg Guler, Michèle Sebag (2026)
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Longitudinal Phase Space Tomography via Machine Learning at the Los Alamos Neutron Science Center
Christopher Leon, Alexander Scheinker, Petr M. Anisimov (2026)
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HELIX: a hybrid envelope-multiparticle linac code with differentiable space-charge optimization
Abhishek Pathak (2026)
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Compact and efficient linear accelerator phase space modeling via implicit neural representation learning
Veng Jean Heng, Serdar Charyyev, Yong Yang, Cynthia Chuang, Lei Xing, Lianli Liu (2026)
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Flexible Spectral-Normalized Neural Gaussian Process for Dynamic Aperture Prediction
Yousra El-Bachir, Frederik Van der Veken, Davide di Croce, Carlo Emilio Montanari, Massimo Giovannozzi, Ekaterina Krymova, Tatiana Pieloni (2026)
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Likelihood-Based Unsupervised Anomaly Detection in CMS Dijet Events
Bhavishya Chebrolu, Hitesh Rasineni, Prajwal Aaryan Immadi (2026)
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Transverse beam profile reconstruction using a hybrid neural network framework for wall current monitor at LEHIPA
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Physics-Informed Drift Diagnosis for Laser-Plasma Accelerator Operations
Ou Labun, Calin Hojbota, Mara Klebonas, Mike Downer, Rafal Zgadza, Phil Franke, Lance Labun (2026)
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Steady Spills, Stronger Signals: Machine Learning for Slow Spill Analysis
Matthew Prescott (2026)
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Strategy of AI Application and Current Activities at European Spallation Source Proton Linac
Ishkhan Gorgisyan, T.J. Shea, Mamad Eshraqi, Karin Rathsmann, Rafael Montaña, Mehdi Mohammednezhad, Hinko Kocevar, Philipp Arnold, Håkan Danared, M.J. Ferreira, P. Pierini, Morten Rostrup Forup Jensen, John G. Weisend, Kent Gustafsson, Gábor Perák, Domen Cvenkel, Karlis Berkolds, Maciej Grzegorzówka, Ciprian Plostinar (2026)
EPJ Research Infrastructures

AI-driven Neutrino Beam Diagnostics for Next-Generation Neutrino Experiments
Noah Aney, U. Chicago (main), Maxwell Chen, Sudeshna Ganguly (2026)
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Similarity Pairing with Energy Mover's Distance for Self-Supervised Pre-Training at the LHC
Ho Fung Tsoi, Dylan Rankin (2026)

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Self-Supervised Learning for Robust Resonance Mass Regression in Cascade Decays

Ho Fung Tsoi, Alex Yang, Luis Felipe Gutierrez Zagazeta, Shion Chen, Dylan Rankin (2026)

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Moderator Modeling for High Intensity Slow Positron Sources

Sophie Crisp, Ryland Goldman, Spencer Gessner (2026)

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Bayesian Optimization of The Relativistic Heavy Ion Collider Luminosity via $\mathcal{S}^{\wedge*}\mathcal{S}$ Control

Xiaofeng Gu, Benjamin Coe, Guillaume Robert-Demolaize, Takeshi Kanesue, Masahiro Okamura, William Fung, Yue Hao, Ji Qiang, Xiaoye S. Li, Yang Liu (2026)

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Design and Operation of a 3-Crystal Multi-Volume Reflection Array for Loss Reduction in CERN SPS Slow Extraction

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